AI-Based Spacecraft Anomaly Detection Using Multivariate Telemetry Time Series

A Project Report submitted by

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in partial fulfillment of the requirements for the award of the degree of

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Department of Computer Science and Engineering

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Declaration

I hereby declare that the work presented in this Project Report titled **AI-Based Spacecraft Anomaly Detection Using Multivariate Telemetry Time Series** submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of M.Sc./M.Sc.-M.Tech./M.Tech. and M.Tech.-Ph.D., is a bonafide record of the research work carried out under the supervision of Dr. Shaily Verma. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

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Certificate

This is to certify that the Project Report titled **AI-Based Spacecraft Anomaly Detection Using Multivariate Telemetry Time Series**, submitted by **Vikash Malakar** (M25AI2155) to the Indian Institute of Technology Jodhpur for the award of the degree of M.Sc./M.Sc.-M.Tech./M.Tech. and M.Tech.-Ph.D., is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Dr. Shaily Verma

Abstract

This project investigates AI-based anomaly detection for spacecraft using multivariate telemetry time series, focusing on the shift from static threshold-based methods to deep learning paradigms. Modern space missions generate high-dimensional data, making it essential to capture complex temporal and inter-sensor relationships. This study presents a comparative analysis of three advanced models: the UnSupervised Anomaly Detection (USAD) autoencoder, the Transformer-based TranAD, and the Graph Deviation Network (GDN). These models are evaluated on NASA benchmark datasets to analyze their performance in terms of detection accuracy, interpretability, and computational efficiency. The results indicate that TranAD is highly effective in detecting subtle contextual anomalies, while GDN provides superior interpretability through relational modeling. Additionally, hardware-aware optimizations are explored to enable deployment in resource-constrained environments. This work contributes toward developing autonomous and efficient health monitoring systems for future space missions.

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1 INTRODUCTION AND BACKGROUND

The operational reliability of spacecraft systems is an absolute prerequisite for mission success, as these complex machines must function in a uniquely hostile environment characterized by high-energy radiation, vacuum, and extreme thermal gradients. As of early 2025, the number of active satellites in orbit has reached approximately 13,883—a figure that now exceeds the global fleet of passenger aircraft. This rapid expansion, driven by commercial megaconstellations and high-resolution Earth observation platforms, has created a significant data challenge. Modern missions, such as the NASA-ISRO Synthetic Aperture Radar (NISAR), are expected to generate upwards of 85 terabytes of telemetry and science data daily.

The primary mechanism for spacecraft health monitoring is telemetry—a multivariate time series consisting of thousands of parameters, including voltages, temperatures, currents, and actuator states, which provide the only diagnostic window for ground operators. Traditionally, anomaly detection in this domain has relied on expert-defined, static Out-Of-Limit (OOL) thresholds. However, these manual systems are increasingly inadequate due to the sheer volume of data, the presence of complex nonlinear interdependencies, and the occurrence of contextual anomalies that are not captured by simple thresholding.

In the context of health management, telemetry data can be mathematically represented as a multivariate time series:

$$X = (x_1, x_2, \ldots, x_T) \tag{1}$$

where each observation at time t is a vector:

$$x_t \in R^n \tag{2}$$

representing n distinct sensor measurements.

A robust anomaly detection system must capture both temporal dependencies—how signals evolve over time—and spatial dependencies—the relationships among different telemetry variables. These dependencies are critical for identifying subtle anomalies that may indicate potential system failures.

To address these challenges, recent research has shifted toward deep learning-based Multivariate Time Series Anomaly Detection (MTSAD). These models are typically trained in an unsupervised or semi-supervised manner on nominal system data to learn patterns of normal behavior. Deviations from these learned patterns are then flagged as anomalies. The ultimate goal is to develop intelligent

systems capable of autonomous fault detection, root-cause analysis, and real-time decision support in resource-constrained space environments.

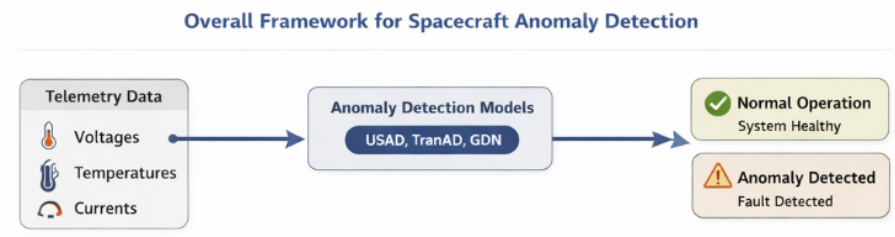

Figure 1: Overall Framework for Spacecraft Anomaly Detection

2 LITERATURE SURVEY

The evolution of spacecraft anomaly detection has progressed from simple statistical techniques to advanced deep learning frameworks capable of modeling complex multivariate telemetry data. Existing approaches can be broadly categorized into multiple generations based on their modeling capabilities.

2.1 Threshold-Based and Initial Recurrent Baselines

Traditional anomaly detection techniques in spacecraft systems primarily rely on Out-Of-Limit (OOL) thresholds and statistical methods. These approaches define safe operating ranges using historical mean and standard deviation values. Any deviation beyond these limits triggers an anomaly alert. While computationally efficient, these methods fail to detect contextual anomalies where values remain within limits but behave abnormally relative to system conditions.

To address these limitations, NASA Jet Propulsion Laboratory introduced the LSTM-NDT framework, which combines Long Short-Term Memory (LSTM) networks with Nonparametric Dynamic Thresholding. LSTMs effectively capture temporal dependencies, while dynamic thresholding adapts to changing system conditions without assuming a fixed distribution.

2.2 Reconstruction-Based Adversarial Models (USAD)

The UnSupervised Anomaly Detection (USAD) model represents a shift toward adversarial deep learning approaches. Standard autoencoders tend to over-generalize, reconstructing both normal and anomalous data effectively. USAD addresses this limitation by employing two decoders in an adversarial setup, forcing the model to amplify reconstruction errors for anomalous inputs.

This dual-decoder mechanism improves anomaly separability while maintaining computational efficiency. USAD is also recognized for its energy-efficient design, making it suitable for large-scale industrial and space applications.

2.3 Transformer-Based Temporal Modeling (TranAD)

TranAD introduces transformer-based architectures for anomaly detection, overcoming limitations of recurrent models such as slow training and difficulty capturing long-range dependencies. By leveraging attention mechanisms, TranAD enables parallel processing of time series data.

A key innovation is the self-conditioning mechanism, where the model performs a two-phase inference. The first phase computes reconstruction error, which is then used to adjust attention weights in the second phase. This approach significantly improves the detection of subtle and contextual anomalies, achieving superior performance compared to traditional models.

2.4 Graph-Based Spatial Modeling (GDN)

The Graph Deviation Network (GDN) focuses on modeling spatial dependencies among telemetry variables. In this approach, each sensor is treated as a node in a graph, and relationships between sensors are learned dynamically.

GDN uses graph attention mechanisms to predict expected sensor behavior based on its neighbors. An anomaly is detected when a sensor deviates from these learned relationships. This method provides enhanced interpretability, enabling effective root-cause diagnosis by identifying which sensor relationships have been disrupted.

2.5 Emerging Trends: KambaAD and Large Language Models

Recent research trends include hybrid state-space models and the integration of Large Language Models (LLMs). KambaAD combines Kolmogorov-Arnold Networks (KAN) with structured state-space models (MAMBA) to improve both coarse-grained and fine-grained anomaly detection.

Additionally, LLM-based systems are being explored for intelligent decision support. These models provide semantic explanations for anomalies, enhancing human interpretability and operational efficiency. Such advancements aim to bridge the gap between predictive accuracy and explainable AI in spacecraft health monitoring.

3 PROBLEM DEFINITION AND OBJECTIVE

3.1 Problem Definition

Spacecraft health monitoring relies on multivariate telemetry data, which can be mathematically defined as a time series:

$$X = \{x_1, x_2, \ldots, x_T\} \tag{3}$$

where each observation is represented as:

$$x_t \in R^n \tag{4}$$

and corresponds to n different sensor measurements such as voltage, temperature, and current.

The objective is to learn a mapping:

$$f : W_t \rightarrow y_t \tag{5}$$

where W_t represents a sliding window of past observations and $y_t \in \{0, 1\}$ indicates whether the system is operating normally or exhibiting anomalous behavior.

Traditional anomaly detection systems face several key challenges:

- **Contextual Deviations:** Sensor values may remain within acceptable limits but behave abnormally under specific operational contexts.
- **Relational Decoupling:** Failures often arise due to disruption in relationships between multiple sensors rather than individual sensor anomalies.
- **High Dimensionality and Noise:** Telemetry data consists of thousands of heterogeneous and noisy channels, making manual monitoring impractical.

3.2 Research Objectives

The primary objective of this research is to evaluate and compare advanced AI-based models for effective spacecraft anomaly detection. The specific objectives are as follows:

- **Evaluate Architectural Trade-offs:** Compare the performance of USAD, TranAD, and GDN models using metrics such as Precision, Recall, and F1-score on NASA benchmark datasets.

- **Analyze Interpretability:** Study how graph-based approaches like GDN assist in identifying root causes of anomalies compared to traditional models.
- **Explore On-Board Feasibility:** Investigate hardware-aware optimization techniques to enable real-time deployment on resource-constrained spacecraft systems.
- **Develop Operational Guidelines:** Propose strategies for selecting appropriate models based on mission requirements such as data availability and computational constraints.

4 METHODOLOGY

The research methodology focuses on a comparative evaluation of three major deep learning architectures—USAD, TranAD, and GDN—using standard NASA telemetry datasets. The workflow includes dataset analysis, preprocessing, model implementation, and optimization for deployment.

4.1 Dataset Analysis (NASA SMAP and MSL)

The primary datasets used in this study are derived from NASA's Soil Moisture Active Passive (SMAP) satellite and Mars Science Laboratory (MSL) rover. These datasets contain real-world labeled anomalies and are widely used for benchmarking anomaly detection models.

The telemetry data is multivariate in nature and includes different categories of parameters:

- **Power/Electrical:** Bus Voltage, HPA Current, Battery State of Charge.
- **Thermal:** Internal temperatures and component temperature readings.
- **Mechanical/ADCS:** Reaction wheel speed and gyroscope measurements.
- **Instrument/Data:** RF power levels and payload sensor outputs.
- **Status/Command:** Discrete operational states and control signals.

The datasets are highly imbalanced, with a large proportion of normal data and relatively few anomaly instances:

- **SMAP:** 55 telemetry channels and multiple anomaly sequences.
- **MSL:** 27 telemetry channels with labeled anomalies.

4.2 Preprocessing and Sliding Window Approach

To prepare the data for model training, a sliding window approach is used. A window of size K is defined as:

$$W_t = \{x_{t-K+1}, \ldots, x_t\} \tag{6}$$

This allows the model to capture temporal dependencies within a fixed-length sequence.

All continuous features are normalized using Min-Max scaling to ensure uniform contribution across different sensor magnitudes.

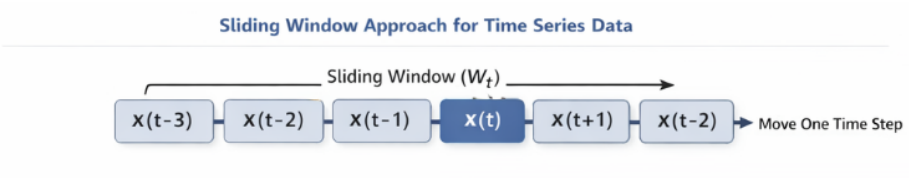

Figure 2: Sliding Window Approach for Time Series Data

4.3 Model Implementation

Three different models are implemented and evaluated:

- **USAD (Adversarial Autoencoder):** Uses an encoder and two decoders to amplify reconstruction errors for anomalous inputs.
- **TranAD (Transformer-based Model):** Utilizes attention mechanisms and a two-phase inference strategy for detecting subtle anomalies.

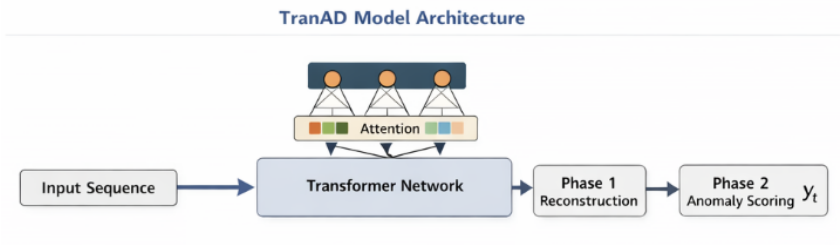

Figure 3: TranAD Model Architecture

- **GDN (Graph Deviation Network):** Models spatial dependencies by representing sensors as nodes in a graph and learning their relationships.

Each model is implemented using PyTorch and evaluated based on its ability to detect anomalies in multivariate time series data.

4.4 Hardware-Aware Optimization

To enable deployment in resource-constrained environments such as spacecraft systems, hardware-aware optimization techniques are applied. Multi-objective optimization is used to balance detection accuracy with computational efficiency.

Optimization techniques aim to reduce:

- Memory usage
- Computational complexity
- Power consumption

This ensures that the proposed models can be deployed on embedded systems such as microcontrollers used in space missions.

5 THEORETICAL, NUMERICAL, AND EXPERIMENTAL FINDINGS

The experimental evaluation of the models—USAD, TranAD, and GDN—was conducted using NASA SMAP and MSL datasets. The results highlight the trade-offs between reconstruction-based, attention-based, and graph-based anomaly detection approaches.

5.1 Comparative Performance Metrics

The models are evaluated using Precision, Recall, and F1-score. The results demonstrate that deep learning approaches significantly outperform traditional methods.

The results indicate that TranAD achieves the highest recall, making it highly effective for detecting subtle anomalies, while GDN provides better precision due to its ability to model inter-sensor relationships.

Table 1: Comparative Performance Results on NASA Benchmarks

Dataset	Model	Precision (%)	Recall (%)	F1-Score (%)
MSL	USAD	88.0	95.3	91.5
MSL	GDN	92.1	91.9	92.0
MSL	TranAD	91.2	97.4	94.2
SMAP	USAD	87.5	98.1	92.4
SMAP	GDN	93.2	93.8	93.5
SMAP	TranAD	94.1	99.8	96.9

Figure 4: Performance Comparison of Models on NASA Datasets

5.2 Analysis of Error Amplification Mechanisms

Each model uses a different strategy to distinguish anomalies:

- **USAD:** Uses adversarial training to amplify reconstruction errors.
- **TranAD:** Employs self-conditioning to focus on subtle deviations.
- **GDN:** Detects anomalies based on disruption in sensor relationships.

These mechanisms improve detection performance by making anomalies more distinguishable from normal data patterns.

5.3 Hardware-Aware Optimization Results

Optimization techniques were applied to reduce computational requirements for deployment on embedded systems.

Table 2: Impact of Model Optimization

Metric	Baseline Model	Optimized Model	Reduction (%)
RAM Usage	2.0 MB	59 KB	97.1
Operations (MACs)	100%	0.6%	99.4
Accuracy	92.7%	88.8%	-3.9

The results demonstrate that significant reductions in memory and computation can be achieved while maintaining acceptable accuracy levels.

5.4 Interpretability and Root-Cause Analysis

Among the evaluated models, GDN provides superior interpretability by identifying relationships between sensors. This enables effective root-cause analysis by highlighting which sensor interactions are disrupted during anomalies.

This feature is particularly useful in mission-critical systems where understanding the source of failure is as important as detecting it.

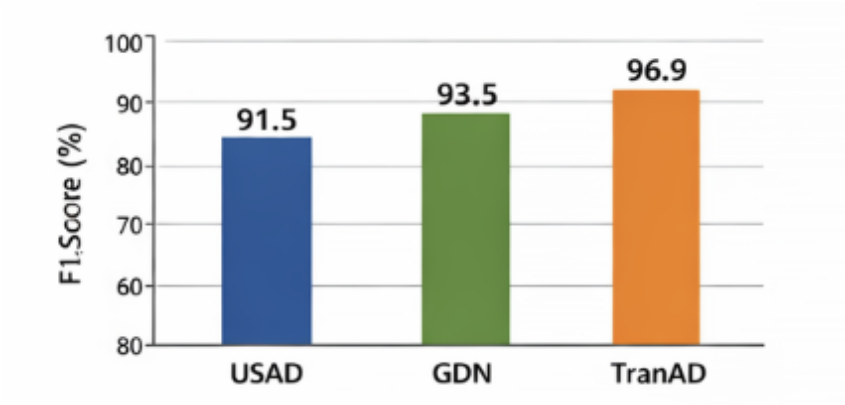

Figure 5: F1 Scores of Each Model

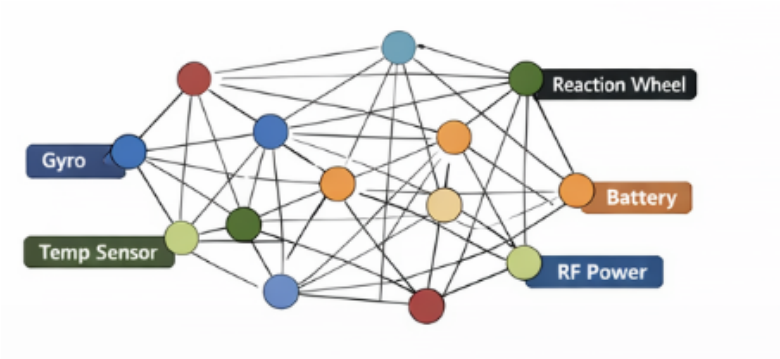

Figure 6: Graph Representation of Sensor Relationships in GDN

6 SUMMARY AND FUTURE PLAN OF WORK

6.1 Summary

This research presents a comprehensive study of AI-based anomaly detection techniques for spacecraft health monitoring using multivariate telemetry time series data. Traditional threshold-based approaches are limited in handling high-dimensional and complex data, motivating the adoption of deep learning-based solutions.

A comparative analysis of three advanced models—USAD, TranAD, and GDN—was conducted using NASA SMAP and MSL datasets. The results demonstrate that TranAD achieves superior performance in detecting subtle and contextual anomalies due to its attention-based architecture. On the other hand, GDN provides enhanced interpretability by modeling relationships between sensors, enabling effective root-cause diagnosis. USAD offers a computationally efficient alternative with stable performance.

Furthermore, hardware-aware optimization techniques were explored to reduce model size and computational requirements. The findings confirm that AI-based anomaly detection can be effectively deployed in resource-constrained environments such as spacecraft systems.

6.2 Future Plan of Work

Building upon the outcomes of this research, several future directions are identified:

- **Integration of State-Space Models:** Investigate advanced architectures such as MAMBA and Kolmogorov-Arnold Networks for efficient long-sequence modeling.
- **Semantic Reasoning with LLMs:** Explore the use of Large Language Models for generating human-readable explanations of anomalies and supporting decision-making.
- **Federated Learning for Space Systems:** Develop distributed learning frameworks that allow multiple spacecraft to collaboratively train models without sharing raw telemetry data.
- **Diffusion-Based Models:** Study the application of diffusion models for improved representation of normal system behavior and enhanced anomaly detection accuracy.

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